Dhruv Nirmal

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PROFESSIONAL SUMMARY

I structure ambiguous commercial problems, turn the analysis into clear recommendations, and stay with the project until those recommendations are implemented. I have done this in hospitality twice over: as the sole analyst running a pricing and margin engagement for an AU/NZ restaurant chain (a product catalog of up to ~3,000 items, category-level price-movement dashboards, supplier deal negotiations, direct reporting to the Chief Procurement Officer and food category manager, around 30% savings identified in their largest category in one year), and as the builder of Facit, a multi-tenant SaaS I designed to give independent cafes a weekly view of food cost %, gross margin, and menu engineering signals, with 14 supplier invoices unified into one model. Both are the same discipline: turning messy, unintegrated hospitality data into a decision a non-financial operator can act on. On top of that, I run analytics for enterprise procurement clients whose combined spend exceeds A$12B, and I have built AI and forecasting workflows in production. Commercial decision support with a genuine technical base, in the room with the people who act on it.

KEY SKILLS

Commercial Analytics: Menu Engineering, Food Cost % / Gross Margin / COGS, Pricing Analysis, GP$ / GP%, Supplier Price-Creep Detection, Promotional Performance, Sales Mix Analysis, Cost-Leakage Identification

BI & Reporting: Power BI, Tableau, Excel (Advanced), Dashboard Design, Scorecards, KPI Reporting, Data Storytelling

Data & Modelling: Python, SQL, ETL Pipelines, Time-Series Forecasting, ML Classification, Anomaly Detection, Statistical Analysis

Stakeholder & Delivery: Client-Facing Delivery, Senior Stakeholder Management, Problem Framing, Cross-functional Communication, Reproducible Processes

EXPERIENCE

Data Analyst, Purchasing Index Data Analytics (Comprara Group)

Jun 2024 to Present | Melbourne, Australia

Sole analyst on multiple enterprise client accounts at a procurement analytics consultancy; the work is commercial decision support delivered directly to CPOs, category managers, and GMs across Australia and New Zealand.

- Owned end-to-end a pricing and margin engagement for an AU/NZ restaurant chain: built the product catalog (up to ~3,000 products), a pricing-decision dashboard tracking category-level price movement to support supplier deal negotiations (potential savings vs. money lost on bad deals, drilled down to product, venue, and supplier), and a supplier dashboard flagging cheaper substitute items. Ran bi-weekly and monthly progress meetings directly with the Chief Procurement Officer, food category manager, and category managers.
- Identified around 30% savings in the client largest category over one year by surfacing price movement and cost leakage at the product and venue level, giving the procurement team a clear basis for negotiating better supplier deals.
- Built an ML spend-categorisation pipeline (TF-IDF, Logistic Regression, one-vs-all binary classifier) across roughly A$12B of client spend, applying a Pareto approach: manual review for the high-value head, model for the long tail. Cut categorisation time from about 1 to 1.5 months of account-manager work to a single-day model run.
- Built 16+ Tableau dashboards and a Power BI solution (adopted by the Department of Transport) giving stakeholder teams live visibility into supplier performance, KPI trends, and cost-optimisation opportunities.

- Developed a time-series forecasting model (Prophet) for a fresh-produce client to plan raw-material and chemical inventory 3 months ahead, incorporating sea-freight price drivers; achieved a 12.5 to 14% error margin, inside the client accepted tolerance.
- Deployed agentic AI workflows (multi-agent data QA, automated client report generation) that cut delivery turnaround by roughly 50% and replaced manual, per-client report-assembly across the team.

Data Engineer Intern, Victorian Centre for Data Insights (VCDI)

Aug 2023 to Nov 2023 | Melbourne, Australia

- Built a Power BI analytics solution adopted by senior Department of Transport stakeholders; presented system architecture and findings to cross-functional teams.
- Engineered a distributed anomaly-detection pipeline (Databricks, PySpark) on government procurement data, improving detection accuracy by about 20%.

Research Data Analyst, Terminal Ballistics Research Laboratory (TBRL), DRDO

Jan 2021 to Jul 2021

- Analysed blast-wave datasets using statistical models and Butterworth filtering; improved noise-level prediction accuracy by about 20% and refined defence-equipment designs by about 10%.

PROJECTS

Facit -- Café Profitability SaaS: Menu Engineering, Food Cost % & Supplier Price-Creep (Python, AI, Multi-source Integration) -- personal build

- Built Facit (myfacit.com) as a white-labellable, multi-tenant SaaS for independent cafes: pulls POS transactions, staff wages, and 14 suppliers' invoices into one unified model so owners know every week whether last week made money, why, and what to change.
- Delivers a weekly profitability pulse covering revenue, wage cost %, food cost %, and gross margin, plus menu engineering signals (which items drive margin, which erode it), supplier price-creep detection (flagging rising input costs before they hit the P&L), and revenue forecasting for the week ahead.
- Built with Neighbours Cafe, St Kilda (a live ~20-staff café) as the design partner: designed around messy, unintegrated real-world data (TapTouch POS, QuickBooks, manual supplier invoices) and a non-technical operator. Running weekly sessions to test and refine the tool in active day-to-day use.

Restaurant Chain Pricing and Margin Platform (Python, SQL, BI Dashboards) -- client work

- Built the end-to-end commercial analytics suite for a multi-site AU/NZ restaurant chain: product catalog (up to ~3,000 items), pricing-decision dashboard tracking category-level price movement for supplier negotiations, and a substitute-product flagging tool. Identified around 30% savings in the client largest category over one year.

ML Spend-Categorisation Pipeline (Python, Scikit-learn, TF-IDF) -- client work

- Led the design and build of a one-vs-all binary ML classifier to categorise roughly A$12B of client spend; applied a Pareto approach (manual head, model tail), handled class imbalance, and packaged the method into a repeatable pipeline now rolled out across clients. Cut categorisation time from about 1 to 1.5 months to a single-day model run.

EDUCATION

Master of Data Science, Monash University | Feb 2022 to Dec 2023

Applied Forecasting, Machine Learning, Statistics I & II, Communicating with Data, Supply Chain Management

Bachelor of Engineering, Thapar University | May 2017 to May 2021

Mathematics I/II/III, Statistics, Entrepreneurship